

Tirzepatide vs Semaglutide

Side-by-side clinical comparison — mechanism, dosing, efficacy, side effects, and patient selection

MECHANISM OF ACTION

SEMAGLUTIDE (OZEMPIC / WEGOVY)	TIRZEPATIDE (MOUNJARO / ZEPBOUND)
GLP-1 receptor agonist ONLY	DUAL GIP + GLP-1 receptor agonist
Reduces appetite via hypothalamic GLP-1 receptors	GIP activates additional fat metabolism pathways
Slows gastric emptying	More powerful appetite suppression via dual CNS action
Stimulates insulin release (glucose-dependent)	Enhanced insulin sensitivity through both pathways
Suppresses glucagon	Suppresses glucagon more completely
Single pathway — one mechanism	Two pathways = additive metabolic effect

CLINICAL EFFICACY COMPARISON

Metric	Semaglutide	Tirzepatide	Source
Average weight loss	~15% body weight	~20–22.5% body weight	STEP-1 vs SURMOUNT-1 trials
% achieving ≥5% loss	87%	91%	Trial data
% achieving ≥15% loss	32%	57%	Trial data — tirzepatide significantly higher
% achieving ≥20% loss	N/A	36%	SURMOUNT-1 (15mg dose)
A1C reduction (T2D)	~1.5–2%	~2–2.5%	Tirzepatide superior for glucose control
Cardiovascular benefit	Proven (SELECT trial)	Emerging data	Semaglutide has more CV outcome data currently

DOSING & TITRATION SCHEDULE

Week	Semaglutide Dose	Tirzepatide Dose	Notes
Weeks 1–4	0.25mg weekly	2.5mg weekly	Starting dose — GI adjustment period
Weeks 5–8	0.5mg weekly	5mg weekly	First dose increase — most patients tolerate well
Weeks 9–12	1mg weekly	7.5mg weekly	Therapeutic range beginning
Weeks 13–16	1.7mg weekly	10mg weekly	Strong appetite suppression established
Weeks 17+	2.4mg weekly (max)	12.5–15mg weekly (max)	Peak dosing — individualize based on response and tolerance

SIDE EFFECT COMPARISON

Side Effect	Semaglutide	Tirzepatide	Management
Nausea	Very common — 44%	Very common — 44%	Small meals, low fat, bedtime injection, antiemetics PRN
Vomiting	24%	25%	Same as nausea. Hydration critical
Diarrhea	30%	23%	BRAT diet, hydration, loperamide PRN
Constipation	24%	35% (higher with tirz)	Fiber, hydration, MiraLax PRN
Injection site rxn	Similar	Similar	Rotate sites. Room temp injection
Thyroid C-cell risk	Black box warning	Black box warning	CI: MTC or MEN2 personal/family history
Pancreatitis	Rare	Rare	CI: history of pancreatitis
Muscle loss	Yes — requires protein	Yes — requires protein	0.8–1g protein/lb IBW daily

PATIENT SELECTION GUIDE		
Patient Profile	Recommended	Rationale
New to GLP-1 therapy, cost-sensitive	Semaglutide	More insurance coverage. More long-term safety data. Proven CV benefit
Maximum weight loss needed	Tirzepatide	22.5% vs 15% average. Dual mechanism more powerful
Failed semaglutide or inadequate response	Tirzepatide	Different mechanism — many patients respond better
Type 2 diabetes — glucose control priority	Tirzepatide	Superior A1C reduction in trials
Known cardiovascular disease	Semaglutide	SELECT trial proved CV mortality benefit (20% reduction)
Cost is primary concern	Semaglutide	Compounded semaglutide widely available and lower cost

KEY CLINICAL REMINDERS
★ Both medications treat a CHRONIC biological condition — weight regain is expected when discontinued. Counsel patients before starting.
★ Protein intake is critical for both — minimum 0.8–1g per pound of ideal body weight daily to prevent muscle loss.
★ Neither medication is approved in: MTC/MEN2 personal or family history, active pancreatitis, pregnancy.
★ Monitor for mood changes — GLP-1 receptors are in the brain. Rare reports of depression and suicidal ideation.
★ Drug interactions: both delay gastric emptying — oral medications (especially thyroid, contraceptives) may have altered absorption.